

Menu Driven Program for Array Operations in C

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1. Aim

To write and execute a menu driven C program that performs the following array operations: (a) counting even, odd, positive, negative and zero elements of an array, (b) searching for an element using the linear search technique, and (c) merging two arrays into a single array.

2. Description / Theory

A menu driven program repeatedly displays a list of choices to the user and performs the operation corresponding to the choice entered, until the user chooses to exit. This program uses a do-while loop together with a switch-case statement to implement the menu, and separate user-defined functions for each operation, which keeps the code modular and easy to read.

- Option 1 uses the modulus operator (%) to classify each element as even or odd, and relational operators to classify it as positive, negative or zero.
- Option 2 implements linear search, which compares the key element with every element of the array one by one until a match is found or the array ends.
- Option 3 copies the elements of two arrays, entered by the user, into a third array to merge them, and then displays all three arrays.
- Option 4 terminates the program.

3. Algorithm

Step 1: Start the program and display the menu with four choices.

Step 2: Read the user's choice.

Step 3: If choice = 1, read n elements into an array, then traverse the array once, incrementing counters for even, odd, positive, negative and zero values, and display the counts.

Step 4: If choice = 2, read n elements into an array and a key value, then traverse the array comparing each element with the key; if a match is found, display its position, otherwise report that the element was not found.

Step 5: If choice = 3, read two arrays of sizes n1 and n2, copy both into a third array of size (n1 + n2), and display all three arrays.

Step 6: If choice = 4, terminate the program.

Step 7: Repeat Steps 2 to 6 until the user selects choice 4.

Step 8: Stop.

4. Source Code

```
/*  
 * Program Title : Menu Driven Array Operations  
 * Description   : This program performs the following operations on
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*           arrays using a menu driven approach:
*           1. Count even, odd, positive, negative and zero
*              elements in an array.
*           2. Search for an element in an array using
*              linear search.
*           3. Merge two arrays into a single array.
*           4. Exit the program.
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*/

#include <stdio.h>

/* Function Declarations */
void countElements(void);
void linearSearch(void);
void mergeArrays(void);
void inputArray(int arr[], int size);
void displayArray(int arr[], int size);

int main()
{
    int choice;

    do
    {
        /* Display Menu */
        printf("\n=====\\n");
        printf("          MENU DRIVEN PROGRAM          \\n");
        printf("=====\\n");
        printf("1. Count Even, Odd, Positive, Negative, Zero Elements\\n");
        printf("2. Search an Element (Linear Search)\\n");
        printf("3. Merge Two Arrays\\n");
        printf("4. Exit\\n");
        printf("=====\\n");
        printf("Enter your choice (1-4): ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                countElements();
                break;

            case 2:
                linearSearch();
                break;

            case 3:
                mergeArrays();
                break;

            case 4:
                printf("\\nExiting the program. Thank you!\\n");
                break;

            default:
                printf("\\nInvalid choice! Please enter a number between 1 and 4.\\n");

```

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        }

        } while (choice != 4);

    return 0;
}

/*
 * Function      : inputArray
 * Description    : Reads 'size' number of integer elements into arr[]
 */
void inputArray(int arr[], int size)
{
    int i;
    for (i = 0; i < size; i++)
    {
        printf("Enter element %d: ", i + 1);
        scanf("%d", &arr[i]);
    }
}

/*
 * Function      : displayArray
 * Description    : Displays all the elements of the array
 */
void displayArray(int arr[], int size)
{
    int i;
    printf("Array elements : ");
    for (i = 0; i < size; i++)
    {
        printf("%d ", arr[i]);
    }
    printf("\n");
}

/*
 * Function      : countElements
 * Description    : Counts even, odd, positive, negative and zero
 *                  elements present in an array
 */
void countElements(void)
{
    int arr[100], n, i;
    int evenCount = 0, oddCount = 0;
    int positiveCount = 0, negativeCount = 0, zeroCount = 0;

    printf("\n--- Count Even, Odd, Positive, Negative, Zero ---\n");
    printf("Enter the number of elements: ");
    scanf("%d", &n);

    inputArray(arr, n);

    for (i = 0; i < n; i++)
    {
        /* Check for even or odd */
        if (arr[i] % 2 == 0)
            evenCount++;
        else

```

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        oddCount++;

        /* Check for positive, negative or zero */
        if (arr[i] > 0)
            positiveCount++;
        else if (arr[i] < 0)
            negativeCount++;
        else
            zeroCount++;
    }

    printf("\n----- Result ----- \n");
    printf("Even numbers      : %d\n", evenCount);
    printf("Odd numbers       : %d\n", oddCount);
    printf("Positive numbers: %d\n", positiveCount);
    printf("Negative numbers: %d\n", negativeCount);
    printf("Zero elements    : %d\n", zeroCount);
}

/*
 * Function      : linearSearch
 * Description    : Searches for a key element in the array using the
 *                  linear search technique
 */
void linearSearch(void)
{
    int arr[100], n, key, i, found = 0, position = -1;

    printf("\n--- Linear Search --- \n");
    printf("Enter the number of elements: ");
    scanf("%d", &n);

    inputArray(arr, n);

    printf("Enter the element to search: ");
    scanf("%d", &key);

    for (i = 0; i < n; i++)
    {
        if (arr[i] == key)
        {
            found = 1;
            position = i;
            break;
        }
    }

    if (found == 1)
        printf("\nElement %d found at position %d (index %d).\n", key, position + 1,
position);
    else
        printf("\nElement %d not found in the array.\n", key);
}

/*
 * Function      : mergeArrays
 * Description    : Merges two arrays entered by the user into a
 *                  single third array and displays it
 */

```

```

void mergeArrays(void)
{
    int arr1[100], arr2[100], merged[200];
    int n1, n2, i, j;

    printf("\n--- Merge Two Arrays ---\n");

    printf("Enter the number of elements in first array: ");
    scanf("%d", &n1);
    printf("Enter elements of first array:\n");
    inputArray(arr1, n1);

    printf("Enter the number of elements in second array: ");
    scanf("%d", &n2);
    printf("Enter elements of second array:\n");
    inputArray(arr2, n2);

    /* Copy first array into merged array */
    for (i = 0; i < n1; i++)
        merged[i] = arr1[i];

    /* Copy second array into merged array */
    for (j = 0; j < n2; j++)
        merged[n1 + j] = arr2[j];

    printf("\nFirst Array  : ");
    displayArray(arr1, n1);

    printf("Second Array : ");
    displayArray(arr2, n2);

    printf("Merged Array : ");
    displayArray(merged, n1 + n2);
}

```

5. Sample Output

The program was compiled successfully using GCC with no errors or warnings, and executed for all four menu options. A sample run is shown below (user input is shown after each prompt):

```

=====
                MENU DRIVEN PROGRAM
=====
1. Count Even, Odd, Positive, Negative, Zero Elements
2. Search an Element (Linear Search)
3. Merge Two Arrays
4. Exit
=====
Enter your choice (1-4): 1

--- Count Even, Odd, Positive, Negative, Zero ---
Enter the number of elements: 5
Enter element 1: 1
Enter element 2: 2
Enter element 3: 3
Enter element 4: -4

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Enter element 5: 0

----- Result -----
Even numbers      : 3
Odd numbers       : 2
Positive numbers: 3
Negative numbers: 1
Zero elements     : 1

Enter your choice (1-4): 2

--- Linear Search ---
Enter the number of elements: 5
Enter element 1: 10
Enter element 2: 20
Enter element 3: 30
Enter element 4: 40
Enter element 5: 50
Enter the element to search: 30

Element 30 found at position 3 (index 2).

Enter your choice (1-4): 3

--- Merge Two Arrays ---
Enter the number of elements in first array: 3
Enter elements of first array:
Enter element 1: 1
Enter element 2: 2
Enter element 3: 3
Enter the number of elements in second array: 2
Enter elements of second array:
Enter element 1: 4
Enter element 2: 5

First Array  : Array elements : 1 2 3
Second Array : Array elements : 4 5
Merged Array : Array elements : 1 2 3 4 5

Enter your choice (1-4): 4

Exiting the program. Thank you!
```

6. Conclusion

The menu driven C program was successfully designed, implemented and tested. It correctly counts even, odd, positive, negative and zero elements of an array, performs linear search to locate a given element, and merges two arrays into a single array, thereby demonstrating the use of loops, conditional statements, arrays and user-defined functions in C.